

Case Study: Streamlining Class Scheduling for Santa Rosa Junior College

By Nima Hejazi (nhejazi@gmail.com)

At Santa Rosa Junior College, scheduling is crucial for smooth operations. District police use schedules to program door locks. Custodians rely on them to clean classrooms and event spaces. Maintenance teams need schedules to address technical requirements.

I was hired as a temporary replacement for Heidi Miranda, an administrative assistant nearing retirement. Heidi had held this position for 16 years. At the start of each semester, she would receive the schedule of all courses and manually input them into an Excel file. At the end of each week, she would use this Excel file, along with the Student Information System (SIS) and the Event Management System (EMS), to create the necessary schedules for the relevant parties. This process was highly labor-intensive and, due to reliance on multiple sources and manual data entry, susceptible to errors.

Understanding the Problem

Soon after I began my work, I realized that the process could be significantly improved with a computerized software solution. However, due to the importance of the documents and the parties involved, it was important to understand who used them and how they were used.

While maintaining the manual process, I began holding meetings with all stakeholders. The output documents were primarily used by two main groups. The first was the district police, who required a schedule detailing the opening and closing times of all campus buildings. They relied on this information to unlock the buildings at the start of the day and secure them after use. They also required a detailed daily schedule of classes, along with a list of events for the day. These two daily documents were essential for maintaining campus security and ensuring they remained informed about ongoing activities.

The second group consisted of custodial and maintenance staff. They also required a daily schedule of classes and a list of events. While they were receiving the same daily schedules and event lists as the district police, I realized their needs were different. Unlike the district police, they needed a schedule that would help them understand what was happening at any given time throughout the day.

The old process involved a single Excel file for the lock schedules, along with seven daily class schedules and a separate list of all events (Figure 1). This process was time consuming, and prone to human error. Since there were multiple sources of truth, it was difficult to determine which source was the most accurate or up-to-date.

Technical Requirements

From a technical standpoint, it was essential to ensure the data remained secure. I chose to keep the data stored locally instead of using a cloud system for two main reasons. First, since the majority of stakeholders were consumers of the exported documents rather than the backend system, direct access to the data wasn't necessary. Second, although SRJC has a programming department, I found it challenging to communicate with them, and I realized they were not willing to cooperate to enable the use of their cloud system and single sign-on features. However, I kept scalability in mind in the process.

The interface of the software was crucial, as I would not be the only one using it. Given that this position required no technical expertise, the interface needed to be simple and user-friendly.

For the reasons I mentioned, I didn't have direct access to the database directly. Although it was possible to view the data in HTML and Excel formats (two different sources, one for classes and the other for events), manually entering the data into the software would have been impractical. Doing so would have only maintained the high intensity and risk of human error.

Final Product

After six weeks, the final product was ready for use. Typically, change is challenging in organizations, so my priority was to ensure the transition was

smooth. The software, which I named Lock Scheduler, is developed using Python, SQLite for the database, and wxPython for the interface. From the end user's perspective, it functions like a regular Windows application (Figure 2).

Data is the key component of this software. It is crucial that users can easily import data from two different sources. Importing the data into the software takes less than 15 seconds, whereas the manual process typically took days. (Figure 3) After testing the result for a few weeks, while maintaining the old process, I made sure the quality of the software was satisfactory.

The exported documents are easier to use for both groups. Rather than Excel format, the documents are exported in PDF format; this ensures maximum portability. (Figure 4) Custodial and maintenance staff receive a table sorted by day, building, class, and time (in that order), allowing them to easily see what's happening at any given moment (Figure 5), whereas the District Police receives daily schedules sorted by start time for each day (Figure 6). Both groups receive a weekly schedule of events on campus as well (Figure 7).

Results

The software delivers significant improvements:

- Scheduling errors have been reduced by 94% (from an average of 18 errors to just 1), largely due to the elimination of human error through automated data importing, duplication detection, and error validation.
- Scheduling time has decreased by an impressive 98% (from an average of 10 hours per week to just 15 minutes), enabling staff to focus on more critical tasks.
- Faculty and administrators have praised the system for its ease of use and reliability.

Figure 1

[illegible]

Weekly Lock Schedule

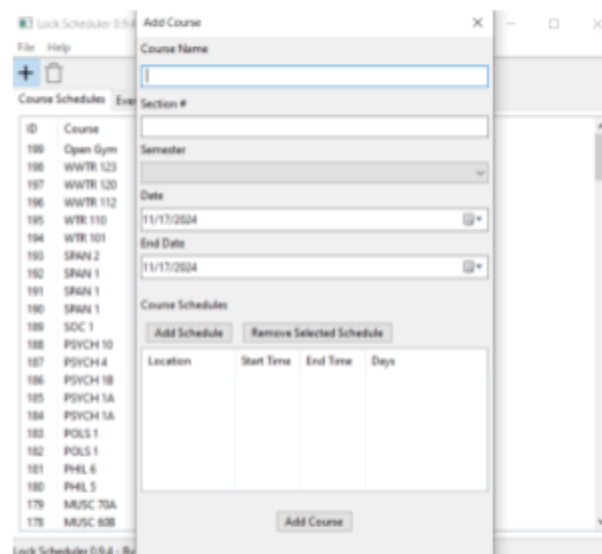
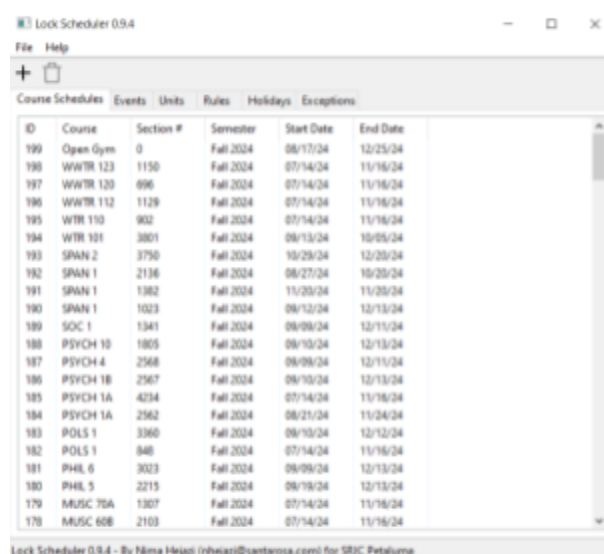
Prattville Event Schedule						
Fall 2024						
Year & Week	RR	RR Date	Event Name	Event Dates	PG	Location
2024-1	RR 10/10/24	4:30pm - 5:30pm	DEAR MY PRINCE	10/10/24 - 10/11/24	PG-13	Lighthouse Theatre
2024-2	RR 10/17/24	4:30pm - 5:30pm	Infantry with the President	10/17/24	TV-14	Ballroom Prisma
2024-3	RR 10/24/24	4:30pm - 5:30pm	From Russia with Love	10/24/24 - 10/25/24	PG-13	Ballroom Prisma
2024-4	RR 10/31/24	4:30pm - 5:30pm	Deadpool & Wolverine	10/31/24	PG-13	Ballroom Prisma
2024-5	RR 11/7/24	4:30pm - 5:30pm	Deadpool & Wolverine	11/7/24 - 11/8/24	PG-13	Ballroom Prisma
2024-6	RR 11/14/24	4:30pm - 5:30pm	Deadpool & Wolverine	11/14/24 - 11/15/24	PG-13	Ballroom Prisma
2024-7	RR 11/21/24	4:30pm - 5:30pm	Deadpool & Wolverine	11/21/24 - 11/22/24	PG-13	Ballroom Prisma
2024-8	RR 11/28/24	4:30pm - 5:30pm	Deadpool & Wolverine	11/28/24 - 11/29/24	PG-13	Ballroom Prisma
2024-9	RR 12/5/24	4:30pm - 5:30pm	Deadpool & Wolverine	12/5/24 - 12/6/24	PG-13	Ballroom Prisma
2024-10	RR 12/12/24	4:30pm - 5:30pm	Deadpool & Wolverine	12/12/24 - 12/13/24	PG-13	Ballroom Prisma
2024-11	RR 12/19/24	4:30pm - 5:30pm	Deadpool & Wolverine	12/19/24 - 12/20/24	PG-13	Ballroom Prisma
2024-12	RR 12/26/24	4:30pm - 5:30pm	Deadpool & Wolverine	12/26/24 - 12/27/24	PG-13	Ballroom Prisma
2024-13	RR 1/2/25	4:30pm - 5:30pm	Deadpool & Wolverine	1/2/25 - 1/3/25	PG-13	Ballroom Prisma
2024-14	RR 1/9/25	4:30pm - 5:30pm	Deadpool & Wolverine	1/9/25 - 1/10/25	PG-13	Ballroom Prisma
2024-15	RR 1/16/25	4:30pm - 5:30pm	Deadpool & Wolverine	1/16/25 - 1/17/25	PG-13	Ballroom Prisma
2024-16	RR 1/23/25	4:30pm - 5:30pm	Deadpool & Wolverine	1/23/25 - 1/24/25	PG-13	Ballroom Prisma
2024-17	RR 1/30/25	4:30pm - 5:30pm	Deadpool & Wolverine	1/30/25 - 1/31/25	PG-13	Ballroom Prisma
2024-18	RR 2/6/25	4:30pm - 5:30pm	Deadpool & Wolverine	2/6/25 - 2/7/25	PG-13	Ballroom Prisma
2024-19	RR 2/13/25	4:30pm - 5:30pm	Deadpool & Wolverine	2/13/25 - 2/14/25	PG-13	Ballroom Prisma
2024-20	RR 2/20/25	4:30pm - 5:30pm	Deadpool & Wolverine	2/20/25 - 2/21/25	PG-13	Ballroom Prisma
2024-21	RR 2/27/25	4:30pm - 5:30pm	Deadpool & Wolverine	2/27/25 - 2/28/25	PG-13	Ballroom Prisma
2024-22	RR 3/6/25	4:30pm - 5:30pm	Deadpool & Wolverine	3/6/25 - 3/7/25	PG-13	Ballroom Prisma
2024-23	RR 3/13/25	4:30pm - 5:30pm	Deadpool & Wolverine	3/13/25 - 3/14/25	PG-13	Ballroom Prisma
2024-24	RR 3/20/25	4:30pm - 5:30pm	Deadpool & Wolverine	3/20/25 - 3/21/25	PG-13	Ballroom Prisma
2024-25	RR 3/27/25	4:30pm - 5:30pm	Deadpool & Wolverine	3/27/25 - 3/28/25	PG-13	Ballroom Prisma
2024-26	RR 4/3/25	4:30pm - 5:30pm	Deadpool & Wolverine	4/3/25 - 4/4/25	PG-13	Ballroom Prisma
2024-27	RR 4/10/25	4:30pm - 5:30pm	Deadpool & Wolverine	4/10/25 - 4/11/25	PG-13	Ballroom Prisma
2024-28	RR 4/17/25	4:30pm - 5:30pm	Deadpool & Wolverine	4/17/25 - 4/18/25	PG-13	Ballroom Prisma
2024-29	RR 4/24/25	4:30pm - 5:30pm	Deadpool & Wolverine	4/24/25 - 4/25/25	PG-13	Ballroom Prisma
2024-30	RR 5/1/25	4:30pm - 5:30pm	Deadpool & Wolverine	5/1/25 - 5/2/25	PG-13	Ballroom Prisma
2024-31	RR 5/8/25	4:30pm - 5:30pm	Deadpool & Wolverine	5/8/25 - 5/9/25	PG-13	Ballroom Prisma
2024-32	RR 5/15/25	4:30pm - 5:30pm	Deadpool & Wolverine	5/15/25 - 5/16/25	PG-13	Ballroom Prisma
2024-33	RR 5/22/25	4:30pm - 5:30pm	Deadpool & Wolverine	5/22/25 - 5/23/25	PG-13	Ballroom Prisma
2024-34	RR 5/29/25	4:30pm - 5:30pm	Deadpool & Wolverine	5/29/25 - 5/30/25	PG-13	Ballroom Prisma
2024-35	RR 6/5/25	4:30pm - 5:30pm	Deadpool & Wolverine	6/5/25 - 6/6/25	PG-13	Ballroom Prisma
2024-36	RR 6/12/25	4:30pm - 5:30pm	Deadpool & Wolverine	6/12/25 - 6/13/25	PG-13	Ballroom Prisma
2024-37	RR 6/19/25	4:30pm - 5:30pm	Deadpool & Wolverine	6/19/25 - 6/20/25	PG-13	Ballroom Prisma
2024-38	RR 6/26/25	4:30pm - 5:30pm	Deadpool & Wolverine	6/26/25 - 6/27/25	PG-13	Ballroom Prisma
2024-39	RR 7/3/25	4:30pm - 5:30pm	Deadpool & Wolverine	7/3/25 - 7/4/25	PG-13	Ballroom Prisma
2024-40	RR 7/10/25	4:30pm - 5:30pm	Deadpool & Wolverine	7/10/25 - 7/11/25	PG-13	Ballroom Prisma
2024-41	RR 7/17/25	4:30pm - 5:30pm	Deadpool & Wolverine	7/17/25 - 7/18/25	PG-13	Ballroom Prisma
2024-42	RR 7/24/25	4:30pm - 5:30pm	Deadpool & Wolverine	7/24/25 - 7/25/25	PG-13	Ballroom Prisma
2024-43	RR 7/31/25	4:30pm - 5:30pm	Deadpool & Wolverine	7/31/25 - 8/1/25	PG-13	Ballroom Prisma
2024-44	RR 8/7/25	4:30pm - 5:30pm	Deadpool & Wolverine	8/7/25 - 8/8/25	PG-13	Ballroom Prisma
2024-45	RR 8/14/25	4:30pm - 5:30pm	Deadpool & Wolverine	8/14/25 - 8/15/25	PG-13	Ballroom Prisma
2024-46	RR 8/21/25	4:30pm - 5:30pm	Deadpool & Wolverine	8/21/25 - 8/22/25	PG-13	Ballroom Prisma
2024-47	RR 8/28/25	4:30pm - 5:30pm	Deadpool & Wolverine	8/28/25 - 8/29/25	PG-13	Ballroom Prisma
2024-48	RR 9/4/25	4:30pm - 5:30pm	Deadpool & Wolverine	9/4/25 - 9/5/25	PG-13	Ballroom Prisma
2024-49	RR 9/11/25	4:30pm - 5:30pm	Deadpool & Wolverine	9/11/25 - 9/12/25	PG-13	Ballroom Prisma
2024-50	RR 9/18/25	4:30pm - 5:30pm	Deadpool & Wolverine	9/18/25 - 9/19/25	PG-13	Ballroom Prisma
2024-51	RR 9/25/25	4:30pm - 5:30pm	Deadpool & Wolverine	9/25/25 - 9/26/25	PG-13	Ballroom Prisma
2024-52	RR 10/2/25	4:30pm - 5:30pm	Deadpool & Wolverine	10/2/25 - 10/3/25	PG-13	Ballroom Prisma

List of Events

August 26, 2024 Monday	
7:30 AM - 5:00 PM	Cafe/Lobby Hrs -- 400 building
8:00 AM - 8:00 PM	Mahoney Library Hrs
8:00 AM - 5:00 PM	Counseling -- Doyle 213/216
8:00 AM - 5:00 PM	Student Success Center -- 500 building PC 114

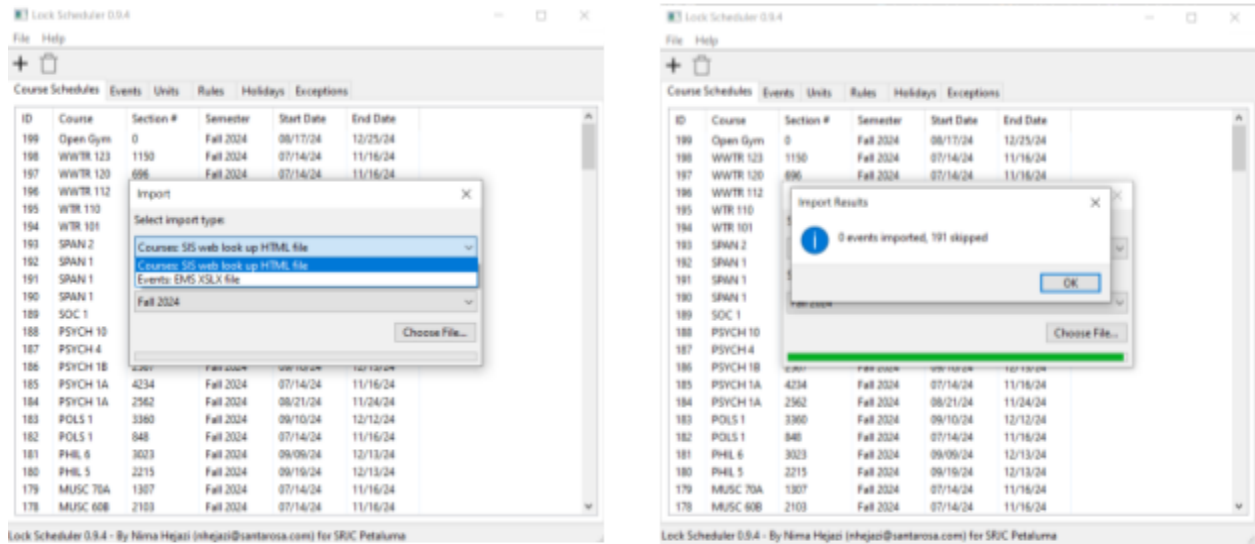
Daily Schedule of Classes

Figure 2



Lock Scheduler's Main Interface

Figure 3



Importing Data into Lock Scheduler

Figure 4

Petaluma Campus Door Lock Schedules 11/18/24 to 11/24/24								
Department/Area	Lock	11/18/2024 Monday	11/19/2024 Tuesday	11/20/2024 Wednesday	11/21/2024 Thursday	11/22/2024 Friday	11/23/2024 Saturday	11/24/2024 Sunday
100 - Inside	PULPL	07:00AM - 10:00PM	07:00AM - 10:00PM	07:00AM - 10:00PM	07:00AM - 10:00PM	07:00AM - 10:00PM	[CLOSED]	[CLOSED]
100 - Inside (Hillside)	PULPL	07:00AM - 10:00PM	07:00AM - 10:00PM	07:00AM - 10:00PM	07:00AM - 10:00PM	07:00AM - 10:00PM	[CLOSED]	[CLOSED]
100 - Inside (Hillside)	PULPL	07:00AM - 10:00PM	07:00AM - 10:00PM	07:00AM - 10:00PM	07:00AM - 10:00PM	07:00AM - 10:00PM	[CLOSED]	[CLOSED]
100 - Inside	PULPL	08:00AM - 05:00PM	08:00AM - 05:00PM	08:00AM - 05:00PM	08:00AM - 05:00PM	08:00AM - 05:00PM	[CLOSED]	[CLOSED]
100 - Inside	PULPL	08:00AM - 05:00PM	08:00AM - 05:00PM	08:00AM - 05:00PM	08:00AM - 05:00PM	08:00AM - 05:00PM	[CLOSED]	[CLOSED]
100 - Inside	PULPL	[CLOSED]	05:00PM - 09:15PM	[CLOSED]	05:00PM - 09:15PM	[CLOSED]	[CLOSED]	[CLOSED]
100 - Inside	PULPL	08:00AM - 05:00PM	08:00AM - 05:00PM	08:00AM - 05:15PM	08:00AM - 05:00PM	08:00AM - 12:00PM	[CLOSED]	[CLOSED]
100 - Inside (Hillside)	PULPL	[CLOSED]	[CLOSED]	[CLOSED]	[CLOSED]	[CLOSED]	[CLOSED]	[CLOSED]
100 - Inside (Hillside)	MULPUL	12:30PM - 03:15PM	09:30AM - 03:45PM	12:30PM - 03:15PM	09:30AM - 03:45PM	[CLOSED]	[CLOSED]	[CLOSED]
100 - Inside (Hillside)	MULPUL	[CLOSED]	[CLOSED]	[CLOSED]	[CLOSED]	[CLOSED]	[CLOSED]	[CLOSED]
100 - Inside	PULPL	[CLOSED]	[CLOSED]	[CLOSED]	[CLOSED]	[CLOSED]	[CLOSED]	[CLOSED]
100 - Inside	PULPL	[CLOSED]	[CLOSED]	[CLOSED]	[CLOSED]	[CLOSED]	[CLOSED]	[CLOSED]
100 - Inside (Hillside)	PULPL	07:00AM - 10:00PM	07:00AM - 10:00PM	07:00AM - 10:00PM	07:00AM - 10:00PM	07:00AM - 10:00PM	[CLOSED]	[CLOSED]
100 - Inside (Hillside)	PULPL	07:00AM - 10:00PM	07:00AM - 10:00PM	07:00AM - 10:00PM	07:00AM - 10:00PM	07:00AM - 10:00PM	[CLOSED]	[CLOSED]
100 - Inside	MULPUL	[CLOSED]	09:30AM - 12:15PM	10:00AM - 03:15PM	09:30AM - 04:15PM	[CLOSED]	[CLOSED]	[CLOSED]
100 - Inside (Hillside)	MULPUL	[CLOSED]	[CLOSED]	[CLOSED]	[CLOSED]	[CLOSED]	[CLOSED]	[CLOSED]
100 - Inside (Hillside)	PULPL	08:00AM - 05:00PM	08:00AM - 05:00PM	08:00AM - 05:00PM	08:00AM - 05:00PM	08:00AM - 01:00PM	[CLOSED]	[CLOSED]
100 - Inside (Hillside)	MULPUL	[CLOSED]	[CLOSED]	[CLOSED]	[CLOSED]	[CLOSED]	[CLOSED]	[CLOSED]
100 - Inside	MULPUL	09:15AM - 04:30PM	09:15AM - 04:00PM	07:15PM - 04:45PM	09:00AM - 05:15AM	[CLOSED]	[CLOSED]	[CLOSED]

Exported from Lock Scheduler version 0.9.4 - Nima Hejazi (nhejazi@santarosa.com)

Lock Schedules Exported by Lock Scheduler

Figure 5

Monday November 11, 2024	
[Veterans Day Holiday]	
Tuesday November 12, 2024	
500 - Call Building Hallway	
PC 639	
01:00 PM - 03:00 PM	ENGL 5
PC 640	
06:00 PM - 08:00 PM	MEDVA 22
08:00 PM - 09:00 PM	MEDVA 123
PC 644	
04:00 PM - 06:00 PM	COJUN 6
06:30 PM - 09:30 PM	EMLS 10
PC 656	
10:00 AM - 12:00 PM	POLS 1
01:00 PM - 03:00 PM	HIST 16.1
PC 657	
10:00 AM - 12:00 PM	HIST 17.1
PC 658	
08:00 AM - 10:00 AM	COMM 1
10:30 AM - 12:00 PM	COMM 1

Daily Schedulers Exported by Lock Scheduler. Used by Custodial and Maintenance Staff.

Figure 6

Monday November 18, 2024		
09:15 AM - 09:00 PM	Open Gym	805 - Physical Ed Weight Room (800 - Physical Ed Building)
09:30 AM - 12:00 PM	ENGL 1A	PC 241 (200 - Doyle Hall 2nd Floor Only)
10:00 AM - 12:00 PM	SOC 1	PC 692 (800 - Call Building Hallway)
12:00 PM - 01:15 PM	MATH 200	PC 696 (800 - Call Building Hallway)
01:00 PM - 03:00 PM	ANTHRD 1L	1200-101 - Anthro Lab (Campus)
01:00 PM - 03:00 PM	BAD 53	252 - Computer Lab (Campus)
01:00 PM - 03:00 PM	BAD 18	PC 693 (800 - Call Building Hallway)
01:00 PM - 03:00 PM	PSYCH 4	PC 697 (800 - Call Building Hallway)
01:30 PM - 03:00 PM	HIST 4.2	PC 656 (800 - Call Building Hallway)
03:00 PM - 05:00 PM	PHIL 6	PC 695 (800 - Call Building Hallway)
04:00 PM - 06:00 PM	COJUN 53	PC 644 (800 - Call Building Hallway)
06:00 PM - 09:30 PM	ADLTED 739	CTC 102 (Campus)
06:30 PM - 09:30 PM	EMLS 771	PC 639 (800 - Call Building Hallway)
06:30 PM - 09:30 PM	EMLS 781	PC 639 (800 - Call Building Hallway)
06:30 PM - 09:45 PM	EMLS 714	PC 644 (800 - Call Building Hallway)
06:30 PM - 09:45 PM	EMLS 713	PC 656 (800 - Call Building Hallway)
06:30 PM - 09:45 PM	EMLS 715	PC 692 (800 - Call Building Hallway)
06:30 PM - 09:45 PM	EMLS 716	PC 692 (800 - Call Building Hallway)

Daily Schedulers Exported by Lock Scheduler. Used by the District Police.

Figure 7

Petaluma Campus Events List from Nov 18 to Nov 24	
Monday November 18, 2024	
SGA Weekly Meetings - Fall24	
Time:02:55 PM - 05:00 PM	
Locations:	
PC 102 (500-101 - Welcome and Connect Center)	
PC 103 (500-101 - Welcome and Connect Center)	
SE 712	
Time:10:00 AM - 02:00 PM	
Locations:	
PC 632 (600 - Call Building Hallway)	

Events List Exported by Lock Scheduler